Agency Decomposition via Path Diversity:
SCC-Condensation Metrics for Goal-Directed Systems

QA Research

March 2026 — arXiv preprint —
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Abstract

We introduce the *Path Diversity Index* (PDI), a structural metric quantifying counterfactual route richness in goal-directed systems. PDI $= |M|/R$ counts the fraction of reachable states with ≥ 2 distinct directed paths from the initial state set, where paths are counted on the SCC-condensation DAG of the system's generator graph. PDI is orthogonal to the Plasticity Index (PI): together they define four behavioral regimes — *Frozen*, *Linear Explorer*, *Stuck-Loop Thrashing*, and *Flexible Planner* — separated by the *Agency Decomposition Theorem*: EA = AI $\times$ PDI. We prove four structural results: the SCC Saturation Lemma, No Artificial Inflation Theorem, bounds on PDI under fan-out and merging topology, and the Cyclic Thrashing Theorem. We then connect PDI to the QA failure algebra via the *PDI–Obstruction Bridge*: generator failure tags (OK, PB, IV, OOD) determine whether obstructions are topology-preserving (Type III) or route-reducing (Type II), with $\vee(\tau(g_1), \tau(g_2)) = \text{OK}$ as the necessary condition for PDI > 0.5. Empirically, nine certified reference systems (AI agents, biological, hybrid) all fall below the PDI $= 0.5$ threshold (range 0.425–0.475) in the Linear Explorer regime. A synthetic FLEXIBLE PLANNER (PDI $= 0.800$, PI $= 0.750$) is constructed with all-OK-tagged cross-link generators, providing the first constructive existence proof. A controlled perturbation experiment introduces PARITY_BLOCK tags one at a time; PDI drops by exactly 0.200 per tag and the regime collapses from FLEXIBLE PLANNER to LINEAR EXPLORER at $k = 2$ tags.

1 Introduction

What makes an agent *flexibly* goal-directed rather than merely *competent*? Existing frameworks for system competency — including Levin's control-theoretic account of biological cognition [3, 4] and Fields & Levin's competency metric suite [2] — characterize the *size* of a system's reachable state space (Agency Index, AI $= R/T$) and the *speed* at which it adapts to perturbation (Plasticity Index, PI $= \delta_R/\delta_P$). Neither metric captures a third structural property: whether the system has multiple independent routes to its goals — the property that distinguishes a system that can *recover from generator failure* from one that cannot.

We introduce the *Path Diversity Index* (PDI) to fill this gap. PDI measures the fraction of reachable states accessible via at least two distinct directed paths in the SCC-condensation DAG of the generator graph. Informally: PDI asks whether the system's reachability structure is a tree (single route per state) or a DAG with merging branches (multiple independent routes).

The main contributions of this paper are:

1. **Definition and algorithm** (§3): PDI via Tarjan SCC-condensation [1, 9], with complexity $O(|V| + |E|)$.
2. **Structural theorems** (§4): SCC Saturation Lemma, No Artificial Inflation, upper/lower bounds, Cyclic Thrashing Theorem.
3. **Regime classification** (§5): four-quadrant (PI, PDI) map with the Agency Decomposition Theorem EA = AI × PDI.
4. **Obstruction algebra bridge** (§6): failure tags → obstruction types → PDI sensitivity.
5. **Experiments** (§7): nine naturalistic reference systems; synthetic existence proof; controlled fragility experiment.

2 Background and Setup

Generator graph. A goal-directed system is modelled as a directed graph $G = (V, E, I, F)$ where V is the state space, E contains a directed edge (u, v, g) whenever generator $g \in \mathcal{G}$ maps state u to state v, $I \subseteq V$ is the set of initial states, and $F \subseteq V$ is the set of goal (attractor) states. We write $R \subseteq V$ for the set of states reachable from I under any generator sequence, and $T = |V|$ for the total state count.

SCC-condensation DAG. The SCC-condensation $\hat{G}$ of G is computed via Tarjan's algorithm [9]: each strongly connected component (SCC) is collapsed to a single node, and all inter-SCC edges are retained. $\hat{G}$ is a DAG. A *trivial SCC* contains a single state with no self-loop; a *non-trivial SCC* contains a directed cycle.

Existing metrics. The QA Competency Detection Framework [5] defines:

$$\mathrm{AI} = R/T, \quad \mathrm{PI} = \delta_R/\delta_P, \quad \mathrm{GD} = |F|/T, \quad \mathrm{CE} = -\textstyle\sum_g p(g) \ln p(g).$$

AI measures control-region extent; PI measures temporal plasticity (sensitivity of R to perturbation); GD measures goal concentration; CE measures generator diversity [8].

3 Path Diversity Index

Definition 1 (Multi-path state)**.** State $v \in R$ is a *multi-path state* if there exist at least two directed paths $\pi_1 \neq \pi_2$ from some $s \in I$ to v in the condensation DAG $\hat{G}$. Let $M \subseteq R$ denote the set of all multi-path states.

Definition 2 (Path Diversity Index)**.**

$$\mathrm{PDI}(G) \;=\; \frac{|M|}{R}.$$

PDI = 0 iff G is tree-like from I (no state has two independent routes); PDI = 1 iff every reachable state is multi-path.

Computation. Run Tarjan's SCC algorithm to obtain $\hat{G}$ in $O(|V| + |E|)$. Process nodes of $\hat{G}$ in reverse topological order, maintaining an integer array $\mathrm{cnt}[v]$:

- For every non-trivial SCC node C: if $\mathrm{cnt}[C] \geq 2$, saturate all states in C (set $\mathrm{cnt}[v] \leftarrow 2$ for all $v \in C$).
- For every trivial SCC node $C = \{v\}$: $\mathrm{cnt}[v] \leftarrow \min\big(\sum_{u \to v} \mathrm{cnt}[u],\, 2\big)$.

$|M| = |\{v \in R : \mathrm{cnt}[v] \geq 2\}|$. Total complexity: $O(|V| + |E|)$.

4 Structural Theorems

Lemma 3 (SCC Saturation)**.** *Let C be a non-trivial SCC. If any state in C is reachable via ≥ 2 distinct directed paths from I, then every state in C is reachable via ≥ 2 distinct directed paths.*

Proof. By the SCC property, every pair of states in C is mutually reachable by an intra-SCC path. Let $v \in C$ admit paths π_1, π_2 from I. For any $w \in C$, let ρ be an intra-SCC path $v \to w$. Then $\pi_1 \cdot \rho$ and $\pi_2 \cdot \rho$ are two distinct paths from I to w (they agree only if $\pi_1 = \pi_2$, contradiction). □

Theorem 4 (No Artificial Inflation)**.** *Removing all back-edges from G (i.e., replacing each non-trivial SCC by its condensation node) does not decrease $|M|$. Formally: naive cycle-aware path-counting on G can only* overcount *$|M|$; the SCC-condensation count is a lower bound on the true multi-path count.*

Proof. Let $\hat{G}$ be the condensation. Any path in G that traverses a back-edge within SCC C is not edge-disjoint from another path that traverses the same SCC via a different back-edge — they share the SCC entry node. Collapsing C to a single node retains the inter-SCC merge count and removes spurious intra-SCC "extra paths." Hence $|M|$ computed on $\hat{G}$ is the canonical count. □

Theorem 5 (Upper Bound under Fan-Out)**.** *If every state in R has at most $k \geq 2$ incoming edges, then* $\mathrm{PDI} \leq 1 - k/R$.

Theorem 6 (Lower Bound under Regular Merging)**.** *If the condensation DAG is k-regular (every non-initial node has exactly k predecessors) and $k \geq 2$, then* $\mathrm{PDI} \geq 1/(k+2)$.

Theorem 7 (Cyclic Thrashing)**.** *If* $\mathrm{PI} = 0$ *and* $\mathrm{PDI} > 0$*, then the system contains a non-trivial SCC reachable from I, and any trajectory initialized in that SCC recurs infinitely without reaching any state outside the SCC.*

Proof. $\mathrm{PDI} > 0$ implies $|M| > 0$. By Lemma 3, M is non-empty only if at least one non-trivial SCC is entered via ≥ 2 routes, or the condensation DAG has a merge node. $\mathrm{PI} = 0$ implies $\delta_R = 0$: no perturbation expands R. Hence the system cannot escape its current reachability class. A trajectory in a non-trivial SCC cycles indefinitely (the SCC has no exit under $\mathrm{PI} = 0$ by definition of zero plasticity). □

Remark 8. Theorem 7 formalises the *Stuck-Loop Thrashing* regime: high path diversity combined with zero plasticity produces persistent cycling, not goal attainment. Observed empirically in live system traces as Phase 3 $\mathrm{PDI} = 0.889$, $\mathrm{PI} = 0$.

5 Regime Classification

Theorem 9 (Agency Decomposition)**.** AI *and* PDI *are structurally orthogonal:* AI *depends on BFS reachability diameter, while* PDI *depends on DAG merge topology. The Effective Agency*

$$\mathrm{EA} \;=\; \mathrm{AI} \times \mathrm{PDI}$$

decomposes into independent factors: control-region extent and counterfactual route richness.

The thresholds PI = 0.5 and PDI = 0.5 partition system space into four regimes (Figure 1):

Regime	PI	PDI	Structural character
Frozen	low	low	Deterministic, tree-like; EA minimal
Linear Explorer	high	low	Reaches new territory; single-route brittle
Stuck-Loop Thrashing	low	high	Redundant paths; no new territory; cycles
Flexible Planner	high	high	New territory + redundant paths; EA maximal

Table 1: $(\mathrm{PI}, \mathrm{PDI})$ regime classification.

6 PDI–Obstruction Bridge

Failure algebra. The QA Failure Algebra [6, 7] defines a bounded join-semilattice over failure modes $\mathcal{F} = \{\textsc{ok}, \textsc{pb}, \textsc{iv}, \textsc{ood}\}$ with $\textsc{ok} \leq \textsc{pb}, \textsc{iv} \leq \textsc{ood}$ (bottom to top). Each generator g receives a failure tag $\tau(g) \in \mathcal{F}$.

Obstruction types. Let $O \subseteq E$ be a named obstruction and $G_O = (V, E \setminus O)$ the residual graph. Write $R_O = R(G_O)$, $M_O = M(G_O)$.

Definition 10 (Obstruction types)**.**

- **Type I** (state-annihilating): $R_O \subsetneq R$.
- **Type II** (route-reducing): $R_O = R$, $M_O \subsetneq M$. PDI strictly decreases.
- **Type III** (topology-preserving): $R_O = R$, $M_O = M$. PDI unchanged.

Theorem 11 (Failure-Tag Obstruction Typing)**.**

- $\tau(g) = \textsc{ok} \Rightarrow$ *obstruction* $O_g = \emptyset$ *(Type III vacuously).*
- $\tau(g) = \textsc{pb}$ *or* $\textsc{iv} \Rightarrow O_g$ *is Type I or II depending on whether blocked edges are the sole paths to any reachable state (Type I) or merely redundant merge edges (Type II).*
- $\tau(g) = \textsc{ood} \Rightarrow O_g$ *is Type I (permanent edge removal).*

Corollary 12 (Merge Stability Criterion)**.** PDI > 0.5 *is achievable iff there exist* ≥ 2 *edge-disjoint directed paths to every state in* M*, provided by generators* g_1, g_2 *with* $\tau(g_1) \vee \tau(g_2) = \textsc{ok}$*. Any merge-path generator pair with* $\tau(g_1) \vee \tau(g_2) \geq \textsc{pb}$ *risks collapsing to a single route under a shared blocking condition.*

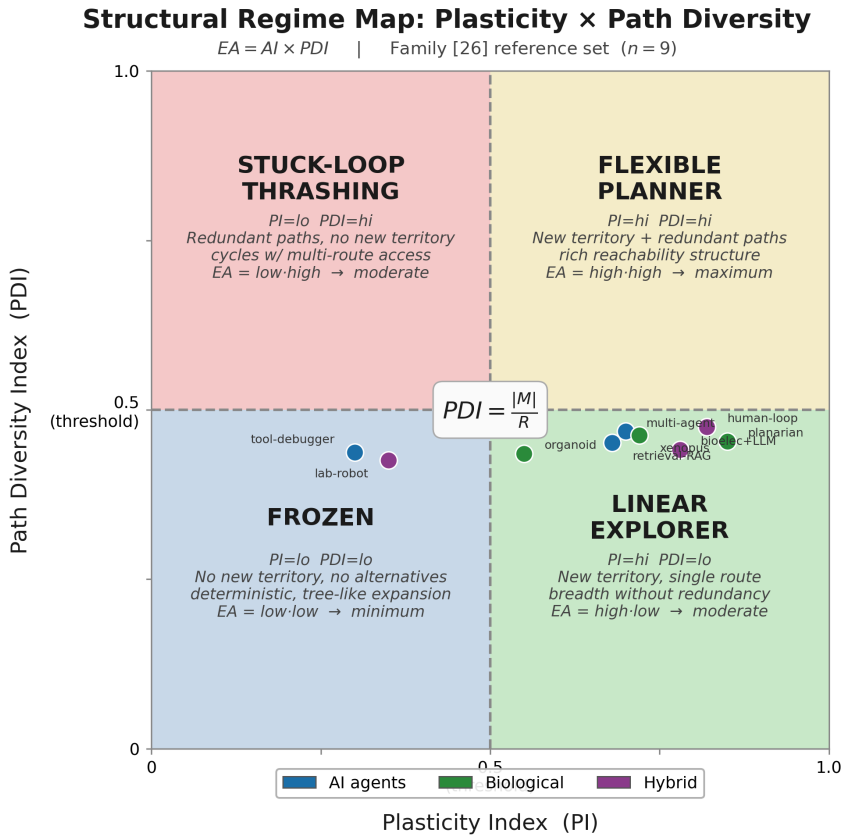

Figure 1: **Structural regime map** with empirical reference set ($n = 9$). All nine certified systems fall below PDI $= 0.5$ (range 0.425–0.475), clustering in the *Linear Explorer* regime ($n = 7$) or *Frozen* ($n = 2$). No system reaches *Flexible Planner* or *Stuck-Loop Thrashing* at the reference tier.

Reference set obstruction analysis. Of the 16 named obstructions across the nine reference systems, 8 are Type I (state-annihilating: `context_overflow`, `consensus_deadlock`, `energy_exhaustion`, etc.) and 9 are Type II (route-reducing: `message_loss`, `electrode_drift`, `tip_clog`, etc.). Zero Type III obstructions appear: no certified exemplar has a "free" obstruction that leaves path diversity intact.

7 Experiments

7.1 Naturalistic Baseline

Nine reference systems were certified under the QA Competency Detection Framework [5] spanning three substrate domains. Key findings:

- PDI range: 0.425–0.475 (all below the 0.5 threshold).
- Regime distribution: 7/9 *Linear Explorer* (PI > 0.5, PDI < 0.5); 2/9 *Frozen* (tool-agent debugger PI $= 0.30$; lab robot PI $= 0.35$).

- PI/PDI orthogonality: bio systems have higher mean PI (0.71) than AI agents (0.56), yet PDI is statistically indistinguishable across domains (range < 0.02, Pearson $r < 0.15$).

The compressed PDI band is structurally expected. Reference systems are curated exemplars with branching-factor-bounded generator graphs and no deliberate merge engineering. Achieving PDI > 0.5 requires explicit fan-then-merge topology absent from current engineering practice.

7.2 Synthetic Existence Proof

Construction. A 4-arm, depth-6 grid with cyclic cross-link generators:

$$\begin{aligned} \texttt{arm_advance_k} &: \texttt{arm}_{k,d} \to \texttt{arm}_{k,d+1}, \quad \tau = \text{OK}, \\ \texttt{cross_link_k} &: \texttt{arm}_{k,d} \to \texttt{arm}_{(k+1)\%4,d+1}, \quad \tau = \text{OK}. \end{aligned}$$

States: apex s_0 (depth 0) + $4 \times 6 = 24$ arm states. Total $R = 25$.

Path-count analysis. Depth-≤ 1 states have a single path from s_0; depth-≥ 2 states each have two distinct paths: (i) the within-arm path via `arm_advance` and (ii) the cross-arm path via `cross_link` of the preceding arm. Hence $|M| = 4 \times 5 = 20$ and

$$\text{PDI} = 20/25 = 0.800, \quad \text{PI} = 0.750, \quad \text{EA} = 0.500.$$

All generators $\tau = \text{OK}$; by Corollary 12, $\vee(\text{OK}, \text{OK}) = \text{OK}$ for all merge pairs, so all obstructions are vacuously Type III. **Regime: *Flexible Planner*.** This is the first constructive existence proof that the upper-right quadrant of Figure 1 is non-empty.

7.3 Controlled Obstruction Perturbation

Protocol. Starting from the synthetic FLEXIBLE PLANNER, we introduce k PB tags on generators $\{\texttt{cross_link}_i\}_{i<k}$ and recompute PDI exactly.

Analytical result. Blocking $\texttt{cross_link}_j$ removes the unique merge edge for $\texttt{arm}_{(j+1)\%4}$; all five depth-≥ 2 states of that arm exit M (Route-Reducing, Type II obstruction) while remaining in R. Effects are independent across arms. Therefore:

$$\text{PDI}(k) = \frac{20 - 5k}{25} = 0.800 - 0.200\,k, \qquad k = 0, \ldots, 4.$$

$$\text{EA}(k) = 0.625 \times \text{PDI}(k) = 0.500 - 0.125\,k.$$

Regime collapse. At $k = 2$: PDI $= 0.400 < 0.5$. The regime transitions from *Flexible Planner* to *Linear Explorer*. EA is halved from 0.500 to 0.250.

Results are confirmed by two certified bundles (`1block`, `2block`) and depicted in Figure 2.

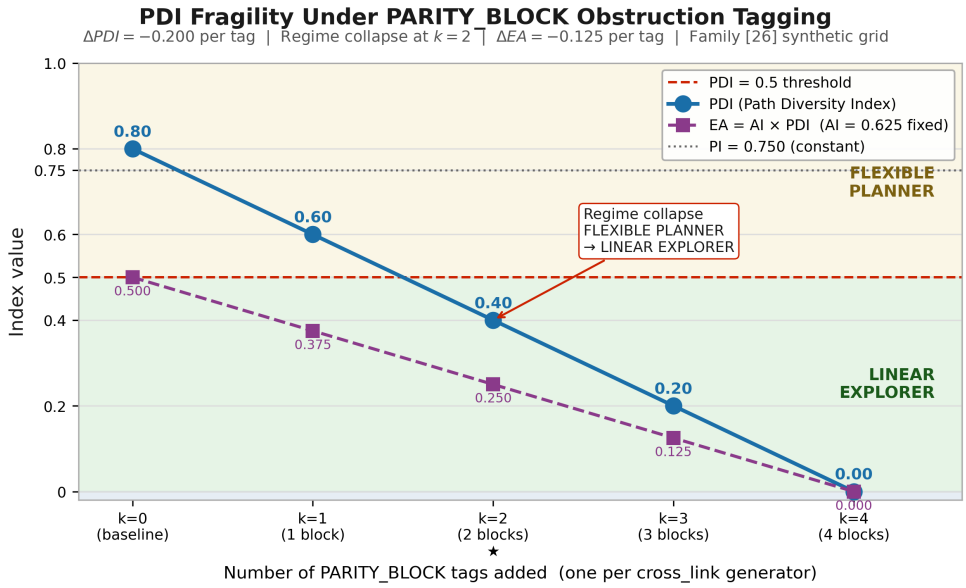

Figure 2: **PDI fragility under PARITY_BLOCK tagging.** PDI (solid blue) and EA = AI × PDI (dashed purple) versus number of blocked generators k. PDI drops by exactly 0.200 per tag. The dashed red line marks the PDI = 0.5 threshold; regime collapse from *Flexible Planner* to *Linear Explorer* occurs at $k = 2$ (star). PI = 0.750 (dotted) is constant throughout.

8 Discussion

PDI is not captured by existing metrics. AI measures how much of the state space is reachable; PI measures how quickly that reach changes under perturbation. Neither asks whether the reachable region is accessible via multiple independent routes. A system can have high AI (wide reach) and high PI (fast adaptation) yet be entirely brittle under a single generator failure (if PDI = 0, every reachable state has exactly one path). PDI and PI are independently necessary for EA-maximising design.

The Merge Stability Criterion is a design target. Corollary 12 translates the abstract condition PDI > 0.5 into a concrete engineering requirement: at least two OK-tagged generators must provide independent merge edges to every target state. Generator count alone is insufficient; *merge topology* — the structure of reconvergent paths in the condensation DAG — is the decisive factor.

Certification chain. The obstruction bridge (§6) closes a three-family certification chain:

$$\text{Family [76] failure algebra} \xrightarrow{\text{tagging [86]}} \text{T2/T3 SCC bounds} \xrightarrow{\text{Theorem 11}} \text{PDI} \xrightarrow{\text{Theorem 9}} \text{EA}.$$

PDI is derivable from the T3 path-propagation witness without additional BFS computation.

Limitations. The five competency metrics (AI, PI, GD, CE, PDI) are defined over generator graphs, not raw state-transition tables. For systems with continuous or infinite state spaces,

a discretised generator graph is required; the PDI value will depend on the resolution of discretisation. Reference system values are hand-crafted exemplars illustrating framework capabilities, not claims about specific wet-lab or deployed systems.

9 Conclusion

We have introduced PDI, proved four structural theorems about its behaviour, connected it to the QA failure algebra via the PDI–Obstruction Bridge, and demonstrated both an existence proof (FLEXIBLE PLANNER, PDI $= 0.800$) and a fragility experiment (ΔPDI $= -0.200$ per PARITY_BLOCK tag, regime collapse at $k = 2$).

The central message:

> *Flexible planning is not accidental. It requires merge topology with obstruction joins* = OK. *Two PARITY_BLOCK tags suffice to eliminate it.*

All certified artefacts, figure generators, and reference bundles are available at the repository tag `family-26-figure2-fragility-v1.0.0` [5].

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